

Irina Tolkova

POSTDOCTORAL RESEARCHER

Cornell K. Lisa Yang Center for Conservation Bioacoustics, Cornell University, Ithaca, NY

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Overview

I am a **postdoctoral researcher** at the **Cornell K. Lisa Yang Center for Conservation Bioacoustics**. My academic path is defined by an interest in **applied mathematics** – with experience across machine learning, signal processing, and robotics – and the commitment to contribute towards bending the arc of biodiversity loss. Now, my goal is to develop computational tools for **rigorous, robust, scalable biodiversity monitoring** to inform **wildlife conservation efforts**, with a focus on acoustic localization and abundance estimation. I'm also interested in broader applications of technology for environmental and social impact.

Professional Positions

Cornell University

EDWARD W. ROSE POSTDOCTORAL FELLOWSHIP

Advised by Dr. Holger Klinck.

Ithaca, NY

2023-

Education

Harvard University

GRADUATE STUDIES

Advised by Prof. L Mahadevan.

- PhD in Applied Mathematics
- MS in Applied Mathematics

Cambridge, MA

2017 - 2023

University of Washington

UNDERGRADUATE STUDIES

- BS in Computer Science with Honors
- BS in Applied and Computational Mathematical Sciences

Seattle, WA

2012 - 2017

Publications

** equal contribution*

PUBLISHED

- Tolkova I**, Klinck H, Cusano DA, Kügler A, & Parks SE (2026). *Detection, communication, and individual identification with deep audio embeddings: A case study with North Atlantic right whales*. PLOS Computational Biology.
- Breviglieri EL, Sugai LSM, Sementili-Cardoso G, Donatelli RJ, & **Tolkova I** (2026). *Traditional bioacoustic analyses and machine-learning methods indicate weak vocal dimorphism in four Cerrado antbird species*. Journal of Ornithology.
- Mohapatra P, **Tolkova I**, Sarkar A, Sehgal A, Salisbury DP, Von Duyke AL, Bouffaut L, Klinck H, Sun JJ (2026). *Agentic Discovery of Multi-Channel Bioacoustic Association Algorithms*. In Foundation Models for Science Workshop at ICLR 2026.
- Dhar N & **Tolkova I** (2025). *Fighting Poaching Through Targeted Deep Learning and Sensor Integration*. In The Thirty-Ninth Annual Conference on Neural Information Processing Systems workshop: AI for non-human animal communication.
- Garcia ML*, **Tolkova I***, Madhusudhana S, Rahaman A, Baker SC, Mayo CA, Hudak CA, Klinck H (2025). *Acoustic abundance estimation for critically endangered North Atlantic right whales in Cape Cod Bay, Massachusetts, USA*. Endangered Species Research.

- Tolkova I** (2023). *Acoustic Source Separation, Contour Classification, and Trajectory Optimization*. Doctoral dissertation, Harvard University.
- Marantan A*, **Tolkova I***, Mahadevan L (2023). *Image cognition using contour curvature statistics*. Proceedings of the Royal Society A, 479(2274), 20220662.
- Swaminathan S*, **Tolkova I***, Baker L, Revi DA, Awad L, Walsh C, Mahadevan L (2022). *A Continuous Statistical-Geometric Framework for Normative and Impaired Gaits*. Journal of the Royal Society Interface.
- Tolkova I**, Klinck H (2022). *Source Separation with an Acoustic Vector-Sensor for Terrestrial Bioacoustics*. The Journal of the Acoustical Society of America, 152(2), 1123-1134.
- Cram DL, van der Wal J, Uomini N, **Tolkova I** (co-author 37/42)... (2022). *The Ecology and Evolution of Human-Wildlife Cooperation*. People and Nature.
- van der Wal J, Spottiswoode C, Uomini N, **Tolkova I** (co-author 38/43)... (2022). *Safeguarding Human-Wildlife Cooperation*. Conservation Letters.
- Chandra J*, Muthupalaniappan S*, Shang Z*, Deng R*, Lin R, **Tolkova I**, Butts D, Sul D, Marzouk S, Bose S, Chen A (2021). *Screening of Parkinson's Disease Using Geometric Features Extracted from Spiral Drawings*. Brain Sciences.
- Tolkova I***, Chu B*, Hedman M*, Kahl S, Klinck H (2021). *Parsing Birdsong with Deep Audio Embeddings*. AI for Social Good Workshop, IJCAI 2021.
- Tolkova I** (2021). *Feature Representations for Conservation Bioacoustics: Review and Discussion*. AI for Social Good Workshop, IJCAI 2021.
- Ciannelli L, **Tolkova I**, Lauth R, Puerta P, Helser T, Gitelman A, Thompson G (2019). *Spatial, Interannual, and Generational Sources of Trait Variability in a Marine Population*. Ecology.
- Torres LG, Orben RA, **Tolkova I**, Thompson DR. (2017) *Classification of Animal Movement Behavior through Residence in Space and Time*. PLOS ONE.

REPORTS

- Klinck, H, Yack T, **Tolkova I**, Salisbury DP, and Harris D (2024). *Passive Acoustic Monitoring in the Mariana Islands: Density Estimation of Beaked Whales along the West Coast of Guam*. Final Report. Prepared for Commander, U.S. Pacific Fleet. Submitted to Naval Facilities Engineering Systems Command Pacific, Pearl Harbor, Hawaii under Contract No. N62470-20-D-0016, Task Order No. 21F0194, issued to HDR Inc., Honolulu, Hawaii.

IN REVIEW

- Smith MA, Cusano DA, Hood D, Robbins J, **Tolkova I**, Weinrich MT, Wiley DN, Parks SE. *Integrating standard identity metrics and machine learning methods to assess acoustic identity encoding in humpback whales (*Megaptera novaeangliae*)*.

IN PREPARATION

- Tolkova I** & Klinck H. *The Cockatiel Party Problem: Source Separation in a Dawn Chorus*.
- Bouffaut L, **Tolkova I**, Pepper BE, Salisbury D, Rahaman A, Ponirakis D, Furtado HSC, Eagen S, Von Duyke AL, Givens GH, Citta JJ, and Klinck H. *Acoustic analysis of the 2025 abundance survey of the Bering-Chukchi-Beaufort bowhead whale population near Point Barrow, Alaska*.
- Tolkova I**, Harris D, Yack T, Salisbury DP, Klinck H. *Diel niche partitioning in passive acoustic monitoring of beaked whales offshore Guam*.
- Akashram & **Tolkova I**. *What do BirdNET Embeddings Capture: The Singer, The Song, or the Setting?*

Presentations

___ presenting author; * equal contribution

INVITED TALKS

- April 2026. Invited speaker for Wildlife Conservation Day at the Cornell College of Veterinary Medicine.
- March 2026. Invited lecturer for Sound, Science, and Society Seminar Course at Cornell University.

October 2025. Invited seminar at Columbia University.

October 2025. Invited speaker for departmental seminar at the Music and Audio Research Laboratory (MARL) at NYU.

July 2025. Virtual lecture for the Ocean Governance and Management course at the University of Aukland.

February 2025. Invited talk for the Museum of the Earth.

May 2024. Scientific Computing and Numerics Seminar at Cornell University.

April 2024. Invited speaker for Brains and Bioacoustics Seminar.

April 2024. Seminar for the Department of Natural Resources at Cornell University.

April 2024. Lunchtime Thursday Seminar at the Cornell Lab of Ornithology.

March 2022. Departmental seminar at the Max Planck Institute for Animal Behavior.

June 2022. IEEE Signal Processing Invited Seminar at the University of Rhode Island.

CONTRIBUTED PRESENTATIONS

Mohapatra P, **Tolkova I**, Sarkar A, Sehgal A, Salisbury DP, Von Duyke AL, Bouffaut L, Klinck H, Sun JJ (2026). *Agentic Discovery of Multi-Channel Bioacoustic Association Algorithms*. Poster presentation at ICLR Foundation Models for Science Workshop (2026)

Akashram*, Bafna A*, Parekh R*, Eric Snyder, Klinck H, **Tolkova I**. *Array geometry evaluation and optimization for bioacoustic localization*. Oral presentation at the Meeting of the Acoustical Society of America (2026)

Citta JJ, Von Duyke AL, Adams B, Stafford KM, Givens GH, **Tolkova I**, Bouffaut L, Klink H, Carroll GM, Kosma M, Phelps JT, Olnes J, Wilford CW, Sformo TL, and Person BT. *Update on the 2025 ice-based bowhead whale abundance survey*. Poster presentation at the Alaska Marine Science Symposium (2026).

Tolkova I. *The Cockatiel Party Problem: Array Processing Challenges in Bioacoustics*. Poster presentation at the Sound and Audio in the Northeast Workshop. (2025)

Garcia LM*, **Tolkova I***, Madhusudhana S, Rahaman A, Baker S, Mayo CA, Hudak CA, Klinck H. *Counting Critically Endangered North Atlantic Right Whales with Acoustics, Airplanes, and AI*. Oral presentation at the Cornell Oceans Symposium (2025)

Tolkova I, Harris D, Yack T, Salisbury DP, Klinck H. *Localization-Derived Acoustic Detection Function for Goose-Beaked Whales Offshore Guam*. Poster presentation at DCLDE Rotterdam (2024)

Tolkova I, Klinck H. *The Cockatiel Party Problem: Separating Sound Sources with an Acoustic Vector Sensor*. Oral presentation at the American Ornithological Society (2024)

Tolkova I & Klinck H. *Source separation with an acoustic vector sensor for terrestrial bioacoustics*. Meeting of the Acoustical Society of America (2023)

Tolkova I & Klinck H. Oral presentation at Northeast Regional Environmental Acoustics Symposium (May 2022)

Tolkova I. *Feature representations for conservation bioacoustics: review and discussion*. Oral presentation at IJCAI AI for Social Good Workshop (August 2021)

Tolkova I. Oral presentation at the UCI CMCF Early Career Researcher Symposium (April 2021)

Tolkova I*, Chu B*, Hedman M*, Kahl S, Klinck H. *Parsing birdsong with Deep Audio Embeddings*. Oral presentation at IJCAI AI for Social Good Workshop (Jan 2021)

Tolkova I*, Bauer L*, Wilby A, Kastner R and Seger K. *Automatic classification of humpback whale social calls*. Oral presentation at the Meeting of the Acoustical Society of America (2017)

Tolkova I, Ciannelli L, Gitelman A, Thompson G, Helser T, Robert L. *Spatial and Temporal Variation in the Size-at-Age of Pacific Cod in the Eastern Bering Sea: Implications for Sampling Strategies*. Poster presentation at International Conference for the Exploration of the Sea (ICES), Copenhagen (2015)

Grants

2025-2026 **co-PI on Point Barrow Bowhead 2025 Whale Census**, North Slope Borough Department of Wildlife Management.

Awards

2025 **Experiential Learning Award**, Cornell Lab of Ornithology
2024 **Experiential Learning Award**, Cornell Lab of Ornithology
2023 **Animal Bioacoustics Best Student Presentation Award**, Acoustical Society of America
2022 **Quantitative Biology Fellowship**, Harvard NSF Simons Center
2021 **Quantitative Biology Fellowship**, Harvard NSF Simons Center
2020 **Quantitative Biology Fellowship**, Harvard NSF Simons Center
2019-2021 **Certificate of Distinction in Teaching**, Harvard Derek Bok Center (awarded 4 times)
2017 **Outstanding Graduating Senior**, Applied Math Department, University of Washington
2015 **Paradise Scholarship**, Robinson Center for Young Scholars, University of Washington
2012-2017 **Annual Dean's List**, University of Washington (awarded 5 times)

Industry Experience

MathWorks

SOFTWARE ENGINEERING INTERN

Summer 2021

Working with the Control Design, Deep Learning, and Reinforcement Learning teams, I prototyped data-driven learning of Koopman embeddings for simulation and control of nonlinear dynamical systems.

Honda Research Institute

RESEARCH INTERN

Summer 2020

As part of HRI's Curious Minded Machines program, I designed and evaluated structured latent representations of high-dimensional environments. Then, I formalized and implemented curious exploration for RL agents (*disentangled VAEs*, *contrastive learning*, *OpenAI Gym*, *PyTorch*, *Stable Baselines*).

Additional Research Experience

Agile Robotics Laboratory (Harvard University)

2017-2019

ADVISED BY PROF. SCOTT KUINDERSMA

- Developed novel non-convex trajectory optimization algorithm (ADMM, augmented Lagrangian methods), benchmarked in simulation for multiple robot platforms (quadrotor, Kuka Arm, RoboBee) against commonly used solvers (SNOPT, IPOPT), and integrated with the Drake robotics toolbox (C++).
- Trained a convolutional denoising autoencoder for signal enhancement of birdsong within outdoor recordings.
- Demonstrated high classification accuracy in training a multilayer perceptron to detect adversarial noise.

Robotics and State Estimation Lab (University of Washington)

2016-2017

ADVISED BY PROF. DIETER FOX AND PROF. BEHCET ACIKMESE

- Constructed data collection framework including point cloud processing and segmentation (ROS, PCL) for graph-based inverse optimal control for learning manipulation tasks from demonstration on the Baxter robot.
- Adapted and demonstrated successive convexification algorithm for real-time trajectory planning for quadrotor drones.

Teaching Experience

Winter 2026	Instructor for Advanced Bioacoustics Workshop , Taught modules on unsupervised machine learning and acoustic localization.	<i>IISER Tirupati</i>
Fall 2023, 2024	Guest speaker for NTRES 7800: Ornithology Seminar , Gave talks about conducting research and pathways in academia.	<i>Cornell</i>
Fall 2024	Lecturer for NTRES 3150: Introduction to Conservation Bioacoustics , Prepared and taught two lectures and labs on machine learning and statistics.	<i>Cornell</i>
Fall 2023	Lecturer for NTRES 3150: Introduction to Conservation Bioacoustics , Prepared and taught four lectures and two labs on machine learning and acoustic localization.	<i>Cornell</i>
Fall 2022	Invited Lecturer for NTRES 3150: Introduction to Conservation Bioacoustics , Prepared and taught an acoustic localization module.	<i>Cornell</i>
Fall 2021	Teaching Fellow for GENED 1080: Engineering the Acoustical World , Led laboratory sessions, developed assignments, held office hours, graded homework.	<i>Harvard</i>
Fall 2020	Head Teaching Fellow for APMTH 22a: Solving and Optimizing , Prepared weekly materials, held office hours, graded homework.	<i>Harvard</i>
Spring 2020	Teaching Fellow for ES 159/259: Introduction to Robotics , Led laboratory sessions, developed assignments, held office hours, graded homework.	<i>Harvard</i>
Fall 2019	Head Teaching Fellow for APMTH 22a: Solving and Optimizing , Prepared weekly materials, taught section, held office hours, graded homework.	<i>Harvard</i>
Fall 2018	Teaching Fellow for CS 182: Introduction to Artificial Intelligence , Prepared weekly section materials, taught section, held exam review and office hours.	<i>Harvard</i>

Mentorship

2025-	Naveen Dhar , <i>High Tech High Mesa, CA</i> . Independent research focused on gunshot detection in rainforest environments; manifested in an oral presentation at ASA (Dec 2025), published conference paper at NeurIPS Workshop on Animal Communication (Dec 2025), press release, and podcast appearance on Scientific American's <i>Science Quickly</i> .
2025-	Rucha Parekh, Arham Bafna , <i>Dwarkadas J. Sanghvi College of Engineering, India</i> . Together with Akashram , this computer science capstone group project focused on array optimization for bioacoustic localization.
2025-	Akashram , <i>Dwarkadas J. Sanghvi College of Engineering, India</i> . Independent research focused on characterizing the effects of various parameters on BirdNET embeddings.
2025	Lillian Wang , <i>Cornell University</i> . Funded 10-week summer research internship focused on exploring techniques for automated acoustic abundance estimation of birds in Sapsucker Woods.
2025	Ivan Mokeyev , <i>Cornell University</i> . Funded 10-week summer research internship focused on designing a board for multichannel audio recording, as well as computationally evaluating the efficacy of different designs for acoustic localization.
2024-	Enrico Breviglieri , <i>São Paulo State University, Brazil</i> . Supported in mentorship on machine learning for an undergraduate thesis on evaluating acoustic sexual dimorphism in Cerrado antbird species; manuscript currently under review.
2024	Ambrose Blay , <i>Cornell University</i> . Funded 10-week summer research internship focused on evaluating acoustic localization with two tetrahedral microphones – including simulation, recording and analysis of natural soundscapes, and controlled playback experiments.

Outreach & Professional Development

PROFESSIONAL SERVICE

- 2025-2026 **Co-chair for Special Session on Machine Learning for Large Passive Acoustic Datasets**, May 2026 Meeting of the Acoustical Society of America
- 2024-2025 **Elected Postdoctoral Representative**, Administrative Board of the Lab of Ornithology
- 2025 **Committee Member**, postdoctoral fellowship search.
- 2024 **Committee Member**, postdoctoral fellowship search.
- 2020-2022 **Lead Organizer for Quantitative Ecology/Ethology/Evolution Seminars**, Coordinated over two dozen virtual talks on a diverse range of topics, with speakers spanning five continents. 
- President of the Harvard GSAS Photography Society**, organized trips, photo competitions, guest speakers, and photographed events and performances for numerous organizations on campus; acquired funding and managed budgets to support these activities. 

VOLUNTEERING AND MENTORSHIP

- 2024- **Volunteer**, Loaves and Fishes, an Ithaca-based community kitchen
- 2024, 2025 **Volunteer**, Migration Celebration, Insectapalooza, and other Lab of Ornithology events
- 2024 **Volunteer speaker**, Skype-A-Scientist (reaching ~100 students across four sessions)
- 2023-2024 **Mentor**, Bioacoustics Equipment and Training program in Indonesia and Malaysia
- 2022 **Mentor**, year-long Veritas AI Bootcamp and Fellowship programs
- 2020-2022 **Mentor**, undergraduate research at the Global Alliance for Medical Innovation
- 2020-2021 **Volunteer tutor**, weekly CovEd program for public schools
- 2018-2019 **Volunteer tutor**, weekly after-school support through Cambridge School Volunteers
- 2013-2017 **Volunteer**, math competitions (GEMS, MathDay, Math Hour Olympiad)

DEVELOPMENT

- 2025 **NSF-Sponsored Workshop on AI, Sensors, and Ecological Infrastructure**, *University of Pittsburgh, PA*. This invited workshop focused on collaborative discussions around the future of artificial intelligence within ecological sensing and biodiversity monitoring.
- 2025 **OOIFB Summer School on Acoustics**, *University of Washington, WA*. This workshop provided an overview of underwater acoustics, including hydrophone networks, seismic sensing, and distributed acoustic sensing.
- 2022 **Graduate Bioacoustics Summer School**, *University of Southern Denmark*. This two-week intensive summer school included lectures, programming exercises, practicals, and experiments spanning diverse topics in bioacoustics.

PEER REVIEW

Nature Scientific Data
Journal of the Acoustical Society of America
Methods in Ecology and Evolution
Ecological Informatics
Marine Environmental Research
Remote Sensing in Ecology and Conservation
Computers and Electronics in Agriculture

HOBBIES

Outside of work, I like to run, explore, and take photos. I also hope to learn documentary filmmaking! 